

HAT-P-19b

R-1474

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_230907 · created 2026-08-01T12:10:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460194.7109 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.047
Transit depth (Rp/Rs)^2	3.15 +/- 1.67 [%]
Orbital Inclination (inc)	85.64 +/- 1.53
Airmass coefficient 1 (a1)	1.014 +/- 0.018
Airmass coefficient 2 (a2)	-0.0098 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	1.81 %
Best Comparison Star	#1 - [243, 229]
Optimal Method	PSF photometry
Transit Duration (day)	0.052 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.178 ± 0.047 vs 0.1418 (deviation 0.77σ)
Duration fit vs published	0.052 d vs 0.1196 d (deviation 1.5σ)
Mid-time O-C	-7.7 min
Depth SNR	3.8
Residual scatter	1.81 %

Light curve

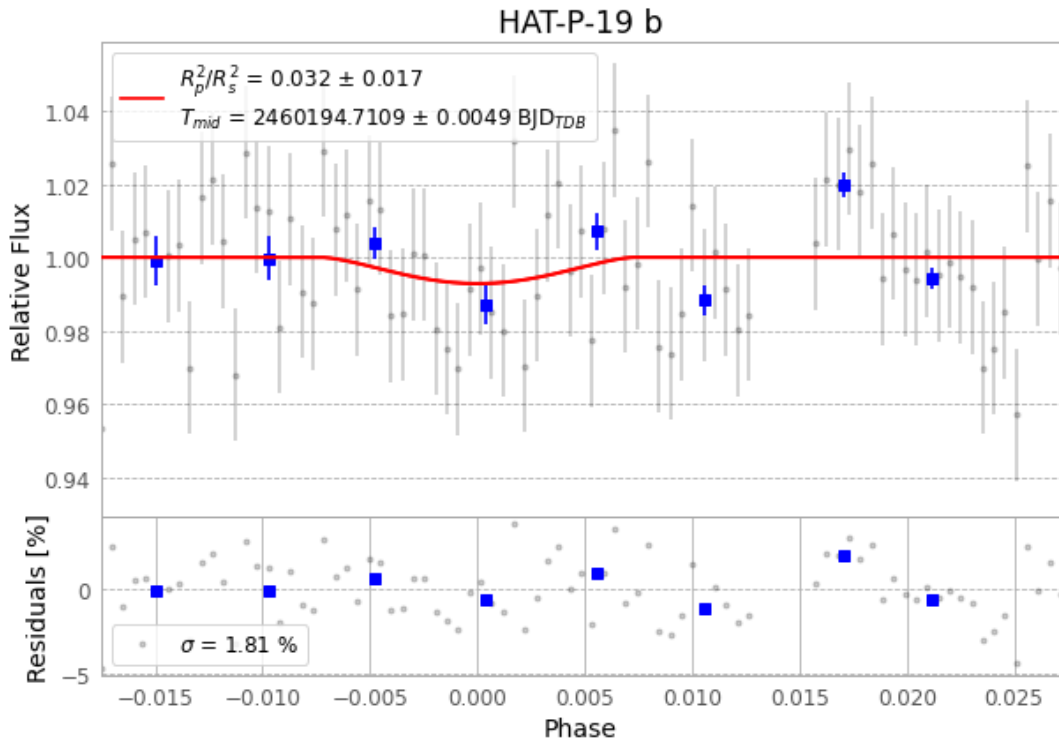


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-09-06.png (SHA-256 ebcd4c8083ecf7b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcd84a0a7c0ed8e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), HATP-19230907031515.FITS → HATP-19230907073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-230907.log (cebe8b1284ec...), FinalLightCurve_HAT-P-19 b_2023-09-06.png (ebcd4c8083ec...), FinalLightCurve_HAT-P-19 b_2023-09-06.csv (ddca4aa05685...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:10Z.